

PROJECT ROADMAP

Kubernetes Autoscaling with Machine Learning Based on Traffic Load Prediction

PROJECT TYPE	ACADEMIC LEVEL	SUPERVISOR
Senior Project	Fourth Year	Dr. Wagdy Refaat Anis

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1. DEVELOPMENT STRATEGY

Development Strategy

Sprint-based execution organized around a project-specific data analytics lifecycle

1-2 WEEKS / SPRINT

EXECUTION MODEL

The project begins with a complete exploratory data analytics lifecycle. The problem is defined from the project statement, then a ready-to-use workload dataset is organized, cleaned, prepared, and explored to identify traffic patterns and useful prediction features. The resulting modeling dataset and EDA findings feed the next phase, where machine learning development and Kubernetes preparation run in parallel before integration, predictive scaling, and evaluation.

01

Project-specific EDA lifecycle

Phase 1 follows six explicit stages from problem definition through validated EDA outputs and feature handoff.

02

Ready-to-use dataset

Traffic data is taken from the selected existing dataset; Kubernetes is not used as the source of training data.

03

Parallel technical development

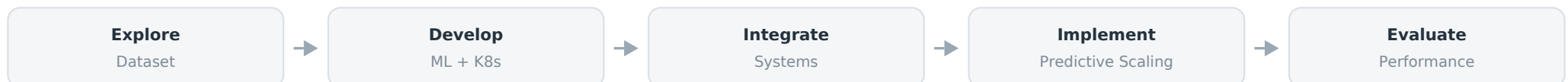
After EDA, ML model development and Kubernetes environment preparation run concurrently.

04

Integration then validation

The model and Kubernetes environment converge for integration, predictive scaling, and controlled evaluation.

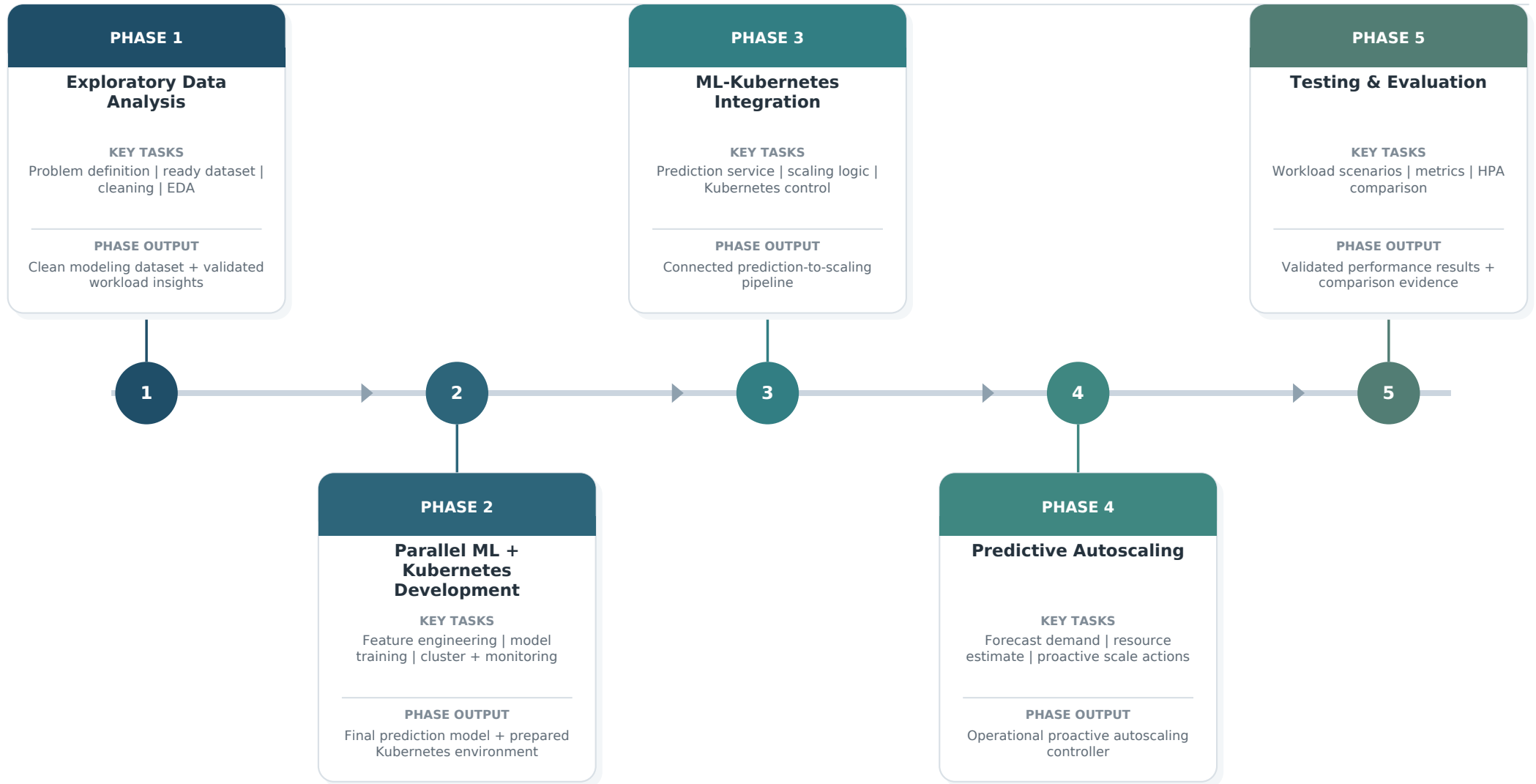
PROJECT FLOW



2. MASTER PROJECT ROADMAP

Master Project Roadmap

Five sequential phases from data understanding to validated predictive autoscaling



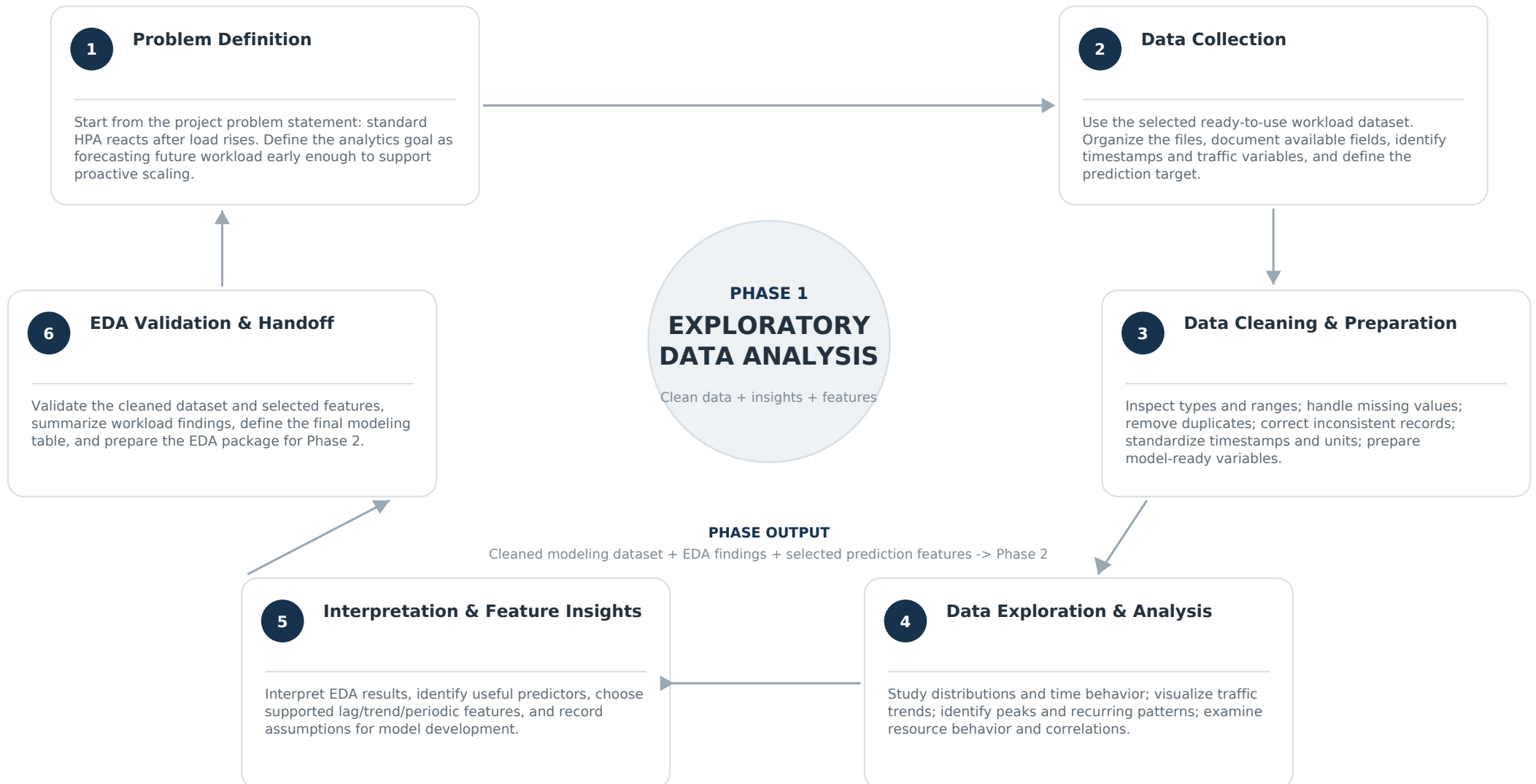
PROJECT PROGRESSION Understand workload -> develop intelligence + platform -> integrate -> automate scaling -> validate performance

3. PHASE 1 ROADMAP

Exploratory Data Analysis

Project-specific data analytics lifecycle - each stage produces the input required by the next stage

SPRINT 1 | 1-2 WEEKS



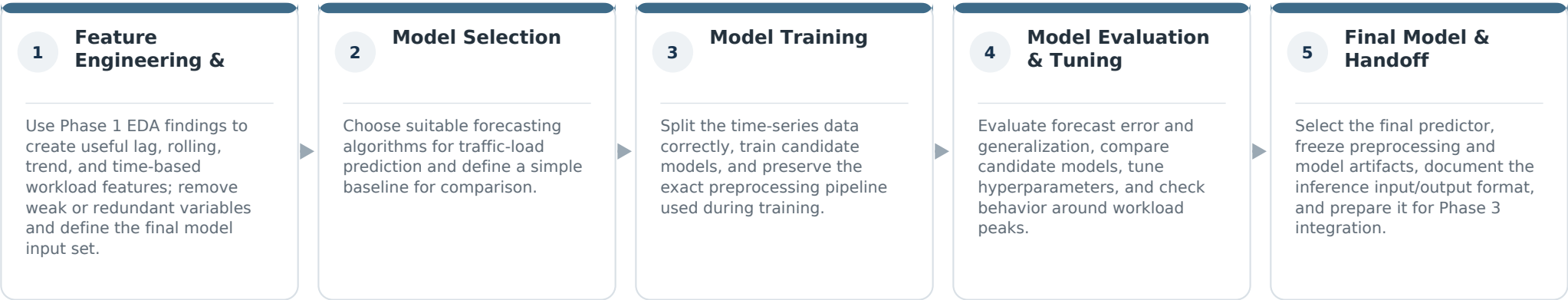
4. PHASE 2 ROADMAP

Parallel ML + Kubernetes Development

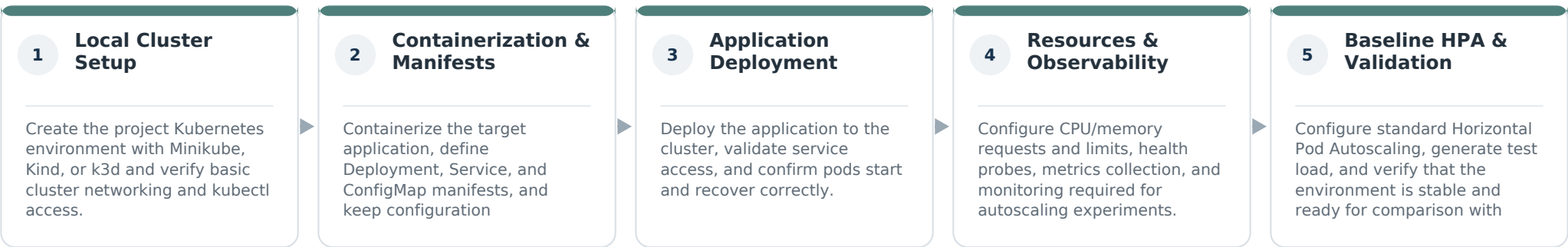
The ML lifecycle resumes after EDA while the Kubernetes environment is built concurrently

SPRINT 2 | 1-2 WEEKS

TRACK A - MACHINE LEARNING LIFECYCLE (CONTINUES FROM PHASE 1 EDA)



TRACK B - KUBERNETES ENVIRONMENT DEVELOPMENT LIFECYCLE



SPRINT OUTPUT
Final traffic prediction model + validated Kubernetes environment with monitoring and baseline HPA.

INPUT TO NEXT PHASE
Phase 3 integrates model forecasts with Kubernetes scaling logic.

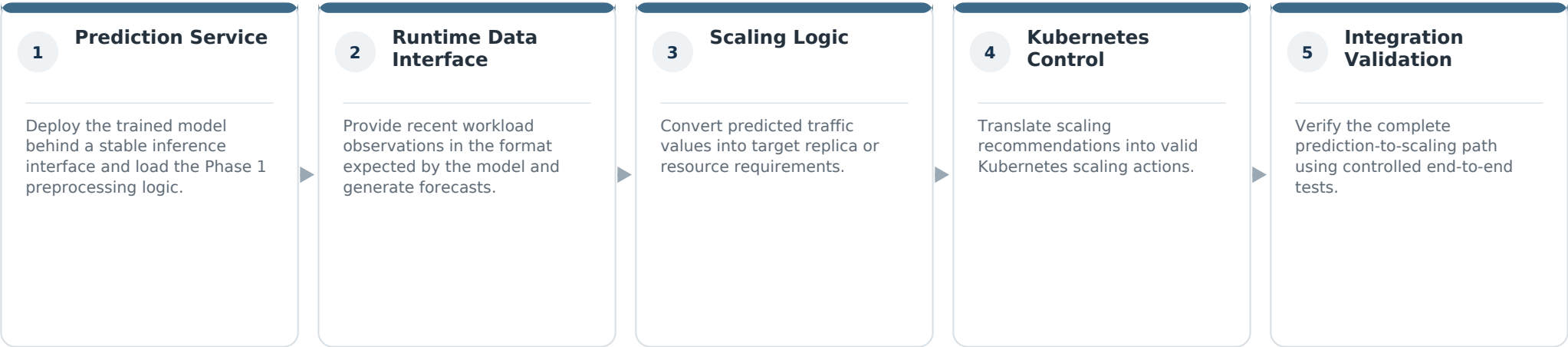
5. PHASE 3 ROADMAP

ML-Kubernetes Integration

Sequential sprint workflow - tasks are completed from left to right

SPRINT 3 | 1-2 WEEKS

TASK SEQUENCE



SPRINT OUTPUT
Connected end-to-end prediction-to-scaling pipeline.

INPUT TO NEXT PHASE
Phase 4 proactive autoscaling behavior and safety mechanisms.

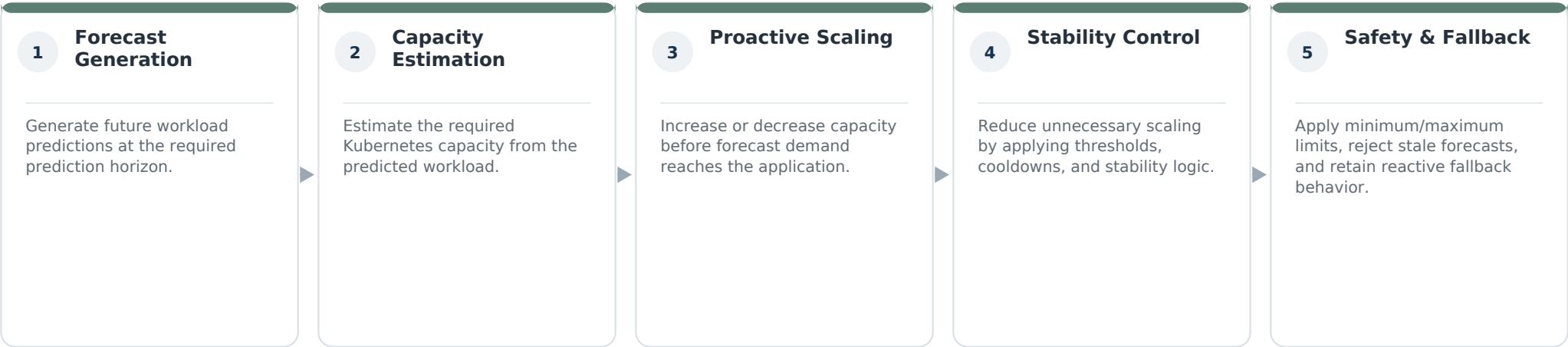
6. PHASE 4 ROADMAP

Predictive Autoscaling Implementation

SPRINT 4 | 1-2 WEEKS

Sequential sprint workflow - tasks are completed from left to right

TASK SEQUENCE



SPRINT OUTPUT
Operational predictive autoscaling controller with safety limits.

INPUT TO NEXT PHASE
Phase 5 controlled testing, evaluation, and comparison.

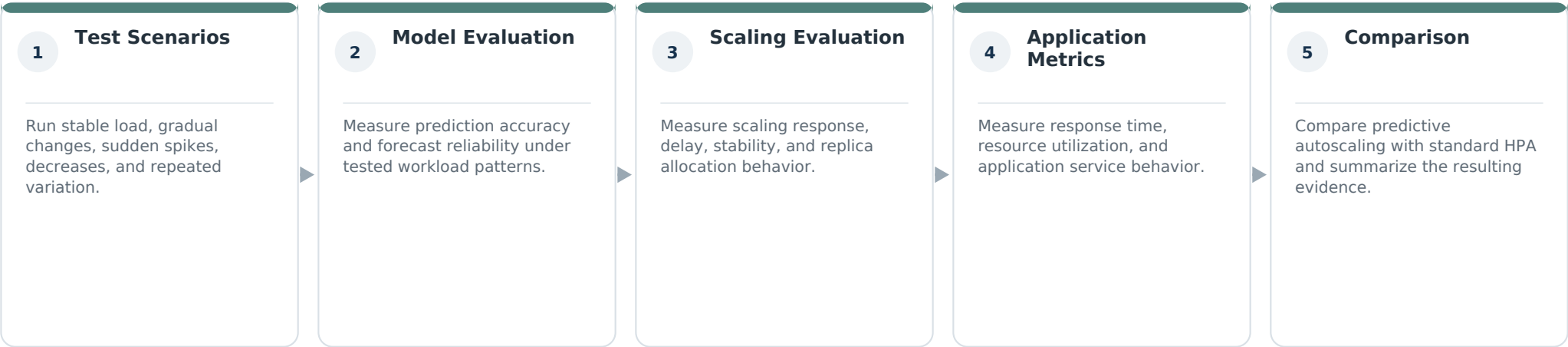
7. PHASE 5 ROADMAP

Testing and Evaluation

Sequential sprint workflow - tasks are completed from left to right

SPRINT 5 | 1-2 WEEKS

TASK SEQUENCE



SPRINT OUTPUT
Validated performance results, comparison evidence, and final system findings.

INPUT TO NEXT PHASE
Final report, demonstration, and project delivery.